REVIEW ARTICLE

Efficacy and safety of tofacitinib for moderate-to-severe plaque psoriasis: a systematic review and meta-analysis of randomized controlled trials

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Abstract

The effects of tofacitinib in treating moderate-to-severe plaque psoriasis were unclear. We aimed to assess the effects of tofacitinib in treating moderate-to-severe plaque psoriasis. We searched PubMed, Cochrane Central Register of Controlled Trials and EMBASE for relevant randomized controlled trials (RCTs) and conducted a systematic review and meta-analysis. Four RCTs with 2724 participants were included. Compared to placebo, tofacitinib significantly improved psoriasis { 75% reduction in the Psoriasis Area and Severity Index score: 5 mg BID: risk difference (RD) 0.32 [95% confidence interval (CI) 0.28–0.35], 10 mg BID: RD 0.51 (95% CI 0.43–0.58); ≥90% reduction in the Psoriasis Area and Severity Index score: 5 mg BID: RD 0.19 (95% CI 0.17–0.22), 10 mg BID: RD 0.36 (95% CI 0.31–0.42); Physician's Global Assessment 0/1: 5 mg BID: RD 0.31 (95% CI 0.27–0.35), 10 mg BID: RD 0.48 (95% CI 0.44–0.53)} and participants' life quality [Dermatology Life Quality Index 0/1: 5 mg BID: RD 0.24 (95% CI 0.20–0.2), 10 mg BID: RD 0.36 (95% CI 0.33–0.40)]. Tofacitinib was associated with an increase in minor adverse events [upper respiratory tract infection: 5 mg BID: RD 0.02 (95% CI 0.00–0.03), 10 mg BID: RD 0.02 (95% CI 0.00–0.04); hypercholesterolaemia: 5 mg BID: RD 0.02 (95% CI 0.01–0.04), 10 mg BID: RD 0.02 (95% CI 0.01–0.04)]. In conclusion, tofacitinib may be a treatment option for moderate-to-severe plaque psoriasis that is unresponsive to other therapies and patients who are intolerable to other therapies or prefer oral medications.

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Conflicts of interest

CK and SW have no conflicts to report; TT received speaking fees from AbbVie Taiwan; CC received speaking fees from AbbVie Taiwan, Janssen-Cilag Taiwan, Novartis Taiwan and Pfizer Taiwan.

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Introduction

Psoriasis is a systemic chronic inflammatory disease that presents with sharply demarcated erythematous scaling plaques.[1–3] Psoriasis has both physical and psychosocial impact on affected patients, including pruritus,[4] vascular inflammation with resultant major adverse cardiovascular events,[5] increased airway inflammation and chronic obstructive pulmonary disease,[6–8] chronic kidney disease,[9] uveitis,[10] impaired life quality,[11] psychiatric disorders,[12] reduce levels of employment and income.[13] Moderate-to-severe plaque psoriasis, defined as Psoriasis Area and Severity Index (PASI) >10 and Dermatology Life Quality Index (DLQI) >10,[14] was estimated to affect 20% of all patients with psoriasis.[15] Patients with moderate-to-severe plaque psoriasis may need treatments with phototherapy, systemic non-biologic medications, or biologics.[15–18] However, concerns about long-term safety and efficacy, for example the association

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between long-term phototherapy and increased risk of skin cancer,[19,20] the relation between prolonged use of systemic non-biologics medications and organ toxicity[16] and the loss of efficacy of biologics,[21] may lead to undertreatment, patients' non-adherence or dissatisfaction with treatments.[22]

Tofacitinib is an oral Janus kinase inhibitor targeting JAK1 and JAK3, thus regulating immune response.[23] Therefore, tofacitinib has been used in treating immune-mediate inflammatory diseases, for example rheumatoid arthritis.[24] Previous studies have found upregulation of JAK3 mRNA expression in psoriatic skin lesions,[25] and overactivation of the JAK/STAT pathway has been implicated in the pathogenesis of psoriasis.[26] Also, an animal experiment has shown that selective inhibition of JAK1/JAK3 signalling reduced psoriasiform inflammation in mice.[27] Therefore, tofacitinib may be used in treating psoriasis based on the ability to interrupt the signalling of cytokines involved in the pathogenesis of psoriasis.[16] The objective of this study was to systematically assess the efficacy and safety of tofacitinib in treating moderate-to-severe plaque psoriasis.

Materials and methods

We conducted a systematic review and meta-analysis of randomized controlled trials (RCTs) on tofacitinib in treating moderate-to-severe plaque psoriasis. We searched the PubMed, Cochrane Central Register of Controlled Trials and EMBASE for relevant RCTs from inception to 6 December 2016. The search strategy was composed of 'psoriasis' and 'tofacitinib/CP690550' with the details listed in Table S1 (Supporting Information). The inclusion criteria of studies were RCTs which evaluated the efficacy and safety of tofacitinib in treating moderate-to-severe plaque psoriasis and provided usable data for assessing the following outcomes. The primary outcomes included the proportion of participants achieving at least 75% reduction in the Psoriasis Area and Severity Index (PASI) score (PASI 75), at least 90% reduction in the PASI score (PASI 90), and Physician's Global Assessment of 'clear' or 'almost clear' (PGA 0/1) at week 12–16. The secondary outcomes were the proportion of participants achieving a Dermatology Life Quality Index score of 0 or 1 (DLQI 0/1) at week 12–16, short-term adverse events (AEs) up to week 12–16, and long-term AEs up to week 52. We specified serious adverse events (SAEs) as AEs that resulted in death, were life-threatening, required hospitalization, resulted in significant disability or resulted in birth defect. Also, we examined specific AEs that the included RCTs commonly reported.

Two authors (CK and CC) independently screened the search results and selected relevant RCTs according to the aforementioned inclusion criteria. One author (CK) extracted the data from the included trials and assessed the risk of bias of included RCTs using the Cochrane Collaboration tool.[28] Another author (CC) verified the extracted data and the evaluation of the risk of bias. Discrepancies were solved by consulting a third author (TT).

We used the Review Manager 5.2 (The Nordic Cochrane Centre, The Cochrane Collaboration, Copenhagen, Denmark, 2012) for executing meta-analysis. We expressed continuous outcomes as risk difference (RD) with 95% confidence interval (CI). We recalculated the extracted data following the intention-to-treat principle, that is we considered those with missing outcome data as treatment failure. The random-effects model was employed in conducting meta-analysis. For significant results, we calculated the corresponding number needed to treat (NNT) and the number needed to treat to harm (NNH) with 95% CI to achieve beneficial and adverse outcomes, respectively. The I^2 statistic was calculated to assess the statistical heterogeneity across the included trials. When the I^2 was over 50%, significant heterogeneity was present.[28]

Results

Characteristics of included studies

The PRISMA study flow chart is shown in Fig. 1. Of 490 records obtained from our search, four RCTs with 2724 participants met our inclusion criteria.[29–31] Table 1 shows the characteristics of the included studies. As to the risk of bias (Fig. 2), three RCTs were rated with a high risk of attrition bias.[29,30]

Efficacy of tofacitinib

Efficacy of tofacitinib in achieving PASI 75 and PASI 90 Three RCTs provided data on both the 5 mg BID and 10 mg BID regimens for the outcomes,[30,31] and one RCT provided only data on the 5 mg BID regimen.[29] As illustrated in Fig. 3, both the 5 mg BID and 10 mg BID groups achieved a significantly higher PASI 75 response than the placebo group [5 mg BID regimen: RD 0.32 (95% CI 0.28–0.35), NNT = 4 (95% CI 3–4); 10 mg BID regimen: RD 0.51 (95% CI 0.43–0.58), NNT = 2 (95% CI 2–3)]. Furthermore, the proportion of achieving PASI response in the 10 mg BID group was significantly higher than the 5 mg BID group ($P < 0.00001$). No statistical heterogeneity was observed across the studies in the 5 mg BID group ($I^2 = 0\%$). However, there was substantial statistical heterogeneity across the studies in the 10 mg BID group ($I^2 = 77\%$).

As illustrated in Fig. 4, the random-effects model meta-analysis revealed a significant increase in achieving PASI 90 response in both the 5 mg BID and 10 mg BID groups when compared to placebo [5 mg BID: RD 0.19 (95% CI 0.17–0.22), NNT = 6 (95% CI 5–6); 10 mg BID: RD 0.36 (95% CI 0.31–0.42), NNT = 3 (95% CI 3–4)]. The proportion of participants achieving PASI 90 in the 10 mg BID group was significantly higher than the 5 mg BID group ($P < 0.00001$). No substantial statistical heterogeneity was observed across the include RCTs (5 mg BID: $I^2 = 0\%$; 10 mg BID: $I^2 = 42\%$).

Efficacy of tofacitinib in achieving PGA 0/1 As aforementioned, one included RCT provided only data on the 5 mg BID

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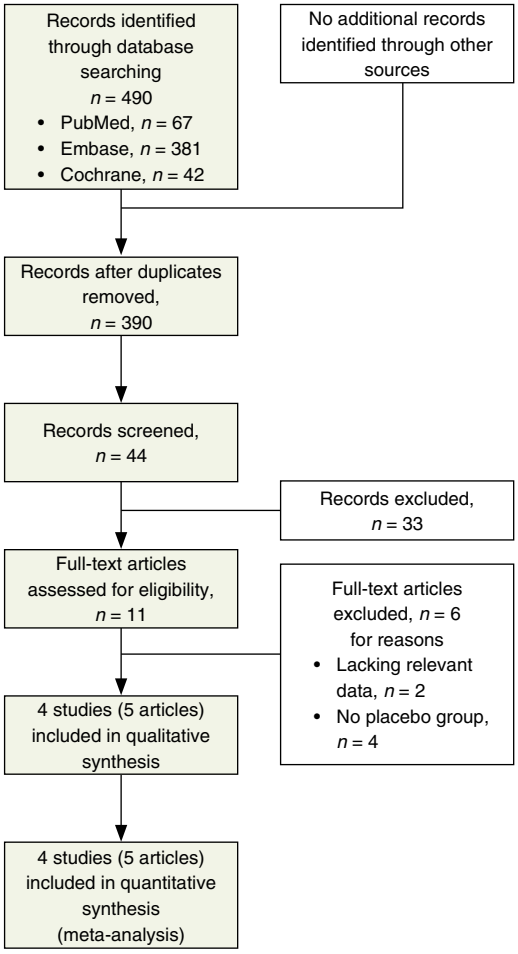

Figure 1 PRISMA study flow chart.

regimen in achieving PGA 0/1,[32] and three other RCTs provided data on this outcome for both 5 mg BID and 10 mg BID regimens.[29,30] As illustrated in Fig. 5, the random-effects model meta-analysis revealed a significant increase in achieving PGA 0/1 in both 5 mg BID and 10 mg BID groups when compared to placebo [5 mg BID: RD 0.31 (95% CI 0.27–0.35), NNT = 4 (95% CI 3–4); 10 mg BID: RD 0.48 (95% CI 0.44–0.53), NNT = 3 (95% CI 2–3)]. The proportion of achieving PGA 0/1 in the 10 mg BID group was significantly higher than the 5 mg BID group ($P < 0.00001$). No statistical heterogeneity was observed across the include RCTs (5 mg BID: $I^2 = 0\%$; 10 mg BID: $I^2 = 20\%$).

Efficacy of tofacitinib for achieving DLQI 0/1 Three RCTs supplied data on both 5 mg BID and 10 mg BID regimens for this outcome.[29,30] One RCT merely provided data on the 5 mg BID regimen.[32] As shown in Fig. S1 (Supporting Information), the random-effects model meta-analysis illustrated a significant increase in achieving DLQI 0/1 in both the 5 mg BID and 10 mg BID groups when compared to placebo [5 mg BID: RD 0.24 (95% CI 0.20–0.28); NNT 5 (95% CI 4–5); 10 mg BID: RD 0.36 (95% CI 0.33–0.40), NNT 3 (95% CI 3–4)]. The proportion of participants achieving DLQI 0/1 was significantly greater in the 10 mg BID group when compared to the 5 mg BID group ($P < 0.00001$). We found no significant statistical heterogeneity across the included RCTs (5 mg BID: $I^2 = 29\%$; 10 mg BID: $I^2 = 0\%$).

Safety of tofacitinib

Short-term AEs Four RCTs provided data concerning this outcome in the 5 mg BID group,[29–31] while three RCTs supplied data for the 10 mg BID group.[29,30] As shown in Fig. S2 (Supporting Information), the short-term AEs did not differ between the 5 mg BID and placebo groups [RD 0.04 (95% CI −0.01 to 0.09)]. The 10 mg BID group was associated with a significant increase in short-term AEs when compared to placebo [RD 0.09 (95% CI 0.04–0.15), NNH 11 (95% CI 6–25)]. There was no statistical heterogeneity across the included RCTs in either the 5 mg BID or 10 mg BID group (both $I^2 = 0\%$).

Short-term SAEs Four RCTs provided data on short-term AEs in the 5 mg BID group,[29–31] while three RCTs supplied data for the 10 mg BID group.[29,30] As shown in Fig. S3 (Supporting Information), the 5 mg BID and placebo groups did not differ in short-term serious AEs [RD 0.01 (95% CI −0.01 to 0.02)]. No did the 10 mg BID and placebo groups [RD 0.00 (95% CI −0.01 to 0.01)]. We found no statistical heterogeneity across the included RCTs in both comparisons (both $I^2 = 0\%$).

Specific AEs All included RCTs reported the following five specific AEs in common: nasopharyngitis, upper respiratory tract infection, hypercholesterolaemia, creatinine phosphokinase (CPK) elevation and headache. Three included RCTs supplied data on these five AEs for both the 5 mg BID and 10 mg BID regimens,[29,30] while one RCT only provided data on the 5 mg BID regimen.[32]

Nasopharyngitis

As in Fig. S4 (Supporting Information), the meta-analysis showed no significant increase of nasopharyngitis in both 5 mg BID and 10 mg BID groups when compared to placebo [5 mg BID: RD −0.03 (95% CI −0.08 to 0.02); 10 mg BID: RD 0.00 (95% CI −0.03 to 0.03)]. A significant statistical heterogeneity was observed in the 5 mg BID group ($I^2 = 65\%$), while no statistical heterogeneity existed in the 10 mg BID group ($I^2 = 4\%$).

Upper respiratory tract infection

As illustrated in Fig. S5 (Supporting Information), the meta-analysis found a significant increase of upper respiratory tract

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Table 1 Characteristics of the included studies

Study	Participants	Intervention	Outcome	Notes
Phase IIb[29,32] (NCT00678210)	• Age: ≥18 years old • Diagnosis: plaque psoriasis for ≥6 months • PASI score: ≥13 • PGA score: 3 ('moderate') or 4 ('severe' • Psoriasis-involved BSA: ≥15% • Candidates for systemic therapy or phototherapy	Randomization carried out by 1 : 1 : 1 : 1 to the followings: • Tofacitinib 2 mg BID for 12 weeks • Tofacitinib 5 mg BID for 12 weeks • Tofacitinib 15 mg BID for 12 weeks • Placebo BID for 12 weeks	PASI 75, PASI 90, PGA 0/1, AEs, SAEs	**Countries**: the USA and Canada **Setting**: 42 centres **Funding**: Pfizer Inc
OPT Pivotal I[30,38] (NCT01276639) and OPT Pivotal II[30,38] (NCT01309737)	• Age: ≥18 years old • Diagnosis: ≥12-month plaque psoriasis • PASI score: ≥12 • PGA score: 3 ('moderate') or 4 ('severe') • Psoriasis-involved BSA: ≥10% • Candidates for systemic therapy or phototherapy	Randomization by 2 : 2 : 1 to the followings: • Tofacitinib 5 mg BID for 52 weeks • Tofacitinib 10 mg BID for 52 weeks • Placebo BID, • Re-randomization to tofacitinib 5 or 10 mg BID at week 16 and continuing to week 52	PASI 75, PASI 90, PGA 0/1, DLQI 0/1, AEs, SAEs	**Countries**: global multicountries **Setting**: 74 (OPT Pivotal I), 94 (OPT Pivotal II) centres **Funding**: Pfizer Inc
OPT Compare[31,39] (NCT01241591)	• Age: ≥18 years old • Diagnosis: plaque psoriasis for ≥12 months • PASI score: ≥12 • PGA score: 3 ('moderate') or 4 ('severe') • Psoriasis-involved BSA: ≥10% • Candidates for systemic therapy or phototherapy • Had failed to respond to, had a contraindication to, or were intolerant to at least one conventional systemic therapy	Randomization carried out by 3 : 3 : 3 : 1 to the following: • Tofacitinib 5 mg BID for 12 weeks • Tofacitinib 10 mg BID for 12 weeks • Etanercept 50 mg twice weekly for 12 weeks • Placebo administered in a matched dosing schedule for 12 weeks	PASI 75, PASI 90, PGA 0/1, DLQI 0/1, AEs, SAEs	**Countries**: global multicountries (excluding the USA and Canada) **Setting**: 122 centres **Funding**: Pfizer Inc

AE, adverse event; DLQI, Dermatology Life Quality Index; PASI, Psoriasis Area and Severity Index; PASI 75, at least 75% reduction in the Psoriasis Area and Severity Index score; PGA, Physician's Global Assessment; SAE, serious adverse event.

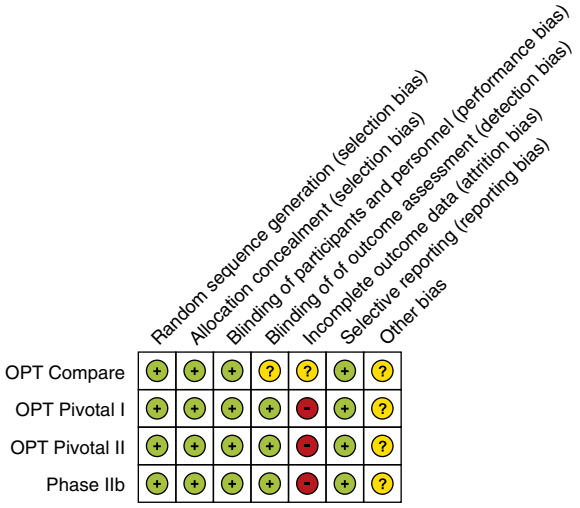

Figure 2 Risk of bias assessment for included studies

infection in both the 5 mg BID and 10 BID groups when compared to placebo [5 mg BID: RD 0.02 (95% CI 0.00–0.03), NNH 50 (95% CI 33 to ∞); 10 mg BID: RD 0.02 (95% CI 0.00–0.04), NNH 50 (95% CI 25 to ∞)]. No significant difference between the 5 mg BID and 10 mg BID groups was found ($P = 0.76$). In both comparisons, there was no statistical heterogeneity across the included RCTs (5 mg BID: $I^2 = 0\%$; 10 mg BID: $I^2 = 32\%$).

Hypercholesterolaemia

In Fig. S6 (Supporting Information), the meta-analysis found a significant increase of hypercholesterolaemia in both the 5 mg BID and 10 mg BID groups [5 mg BID: RD 0.02 (95% CI 0.01–0.04), NNH 50 (95% CI 25–100; 10 mg BID: RD 0.02 (95% CI 0.01–0.04), NNH 50 (95% CI 25–100)]. There was no significant difference in hypercholesterolaemia between the 5 mg BID and 10 mg BID groups. Across the included RCTs, no statistical heterogeneity in both comparisons was observed (both $I^2 = 0\%$).

CPK elevation

As demonstrated in Fig. S7 (Supporting Information), the meta-analysis found no significant increase of CPK elevation in the 5 mg BID group but in the 10 mg BID group [5 mg BID: RD 0.02 (95% CI −0.01 to 0.05); 10 mg BID: RD 0.04 (95% CI

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Study or Subgroup	Tofacitinib Events	Tofacitinib Total	Placebo Events	Placebo Total	Weight	Risk Difference M-H, Random, 95% CI
1.1.1 5 mg bid						
Phase IIb	20	49	1	50	6.0%	0.39 [0.25, 0.53]
OPT Pivotal I	129	363	9	177	35.5%	0.30 [0.25, 0.36]
OPT Pivotal II	145	382	16	196	32.1%	0.30 [0.24, 0.36]
OPT Compare	130	329	6	107	26.3%	0.34 [0.27, 0.41]
Subtotal (95% CI)		**1123**		**530**	**100.0%**	**0.32 [0.28, 0.35]**
Total events	424		32			
Heterogeneity: Tau² = 0.00; Chi² = 1.89, df = 3 (P = 0.60); I² = 0%						
Test for overall effect: Z = 17.66 (P < 0.00001)						
1.1.2 10 mg bid						
OPT Pivotal I	199	360	9	177	34.1%	0.50 [0.44, 0.56]
OPT Pivotal II	200	381	16	196	33.5%	0.44 [0.38, 0.51]
OPT Compare	210	330	6	107	32.4%	0.58 [0.51, 0.65]
Subtotal (95% CI)		**1071**		**480**	**100.0%**	**0.51 [0.43, 0.58]**
Total events	609		31			
Heterogeneity: Tau² = 0.00; Chi² = 8.56, df = 2 (P = 0.01); I² = 77%						
Test for overall effect: Z = 13.05 (P < 0.00001)						

Test for subgroup differences: Chi² = 19.92, df = 1 (P < 0.00001), I² = 95.0%

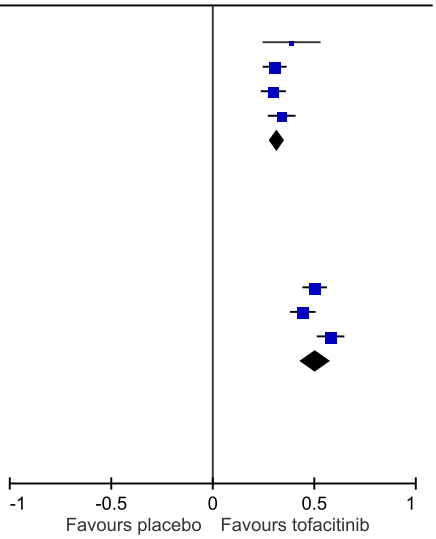

Figure 3 Meta-analysis of efficacy of tofacitinib for patients achieving at least 75% reduction in the Psoriasis Area and Severity Index score.

Study or Subgroup	Tofacitinib Events	Tofacitinib Total	Placebo Events	Placebo Total	Weight	Risk Difference M-H, Random, 95% CI
1.2.1 5 mg bid						
Phase IIb	7	49	0	50	6.6%	0.14 [0.04, 0.25]
OPT Pivotal I	72	363	1	177	38.3%	0.19 [0.15, 0.24]
OPT Pivotal II	94	382	10	196	24.6%	0.20 [0.14, 0.25]
OPT Compare	69	329	1	107	30.5%	0.20 [0.15, 0.25]
Subtotal (95% CI)		**1123**		**530**	**100.0%**	**0.19 [0.17, 0.22]**
Total events	242		12			
Heterogeneity: Tau² = 0.00; Chi² = 1.01, df = 3 (P = 0.80); I² = 0%						
Test for overall effect: Z = 14.33 (P < 0.00001)						
1.2.2 10 mg bid						
OPT Pivotal I	142	360	1	177	53.2%	0.39 [0.34, 0.44]
OPT Pivotal II	148	381	10	196	46.8%	0.34 [0.28, 0.40]
OPT Compare	119	330	1	0		Not estimable
Subtotal (95% CI)		**1071**		**373**	**100.0%**	**0.36 [0.31, 0.42]**
Total events	409		12			
Heterogeneity: Tau² = 0.00; Chi² = 1.73, df = 1 (P = 0.19); I² = 42%						
Test for overall effect: Z = 14.06 (P < 0.00001)						

Test for subgroup differences: Chi² = 34.86, df = 1 (P < 0.00001), I² = 97.1%

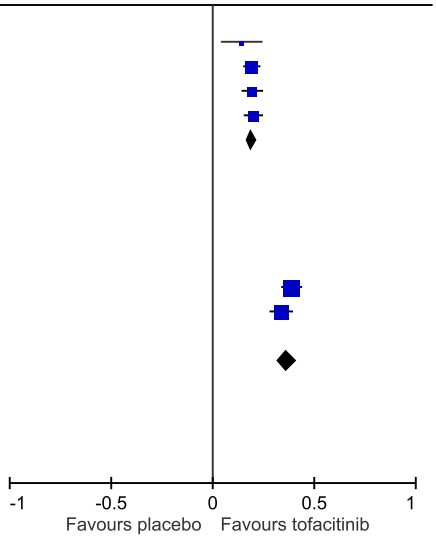

Figure 4 Meta-analysis of efficacy of tofacitinib for patients achieving at least 90% reduction in the Psoriasis Area and Severity Index score.

0.01–0.06), NNH 25 (95% CI 16–100)]. Significant statistical heterogeneity was present in the 5 mg BID group (I^2 = 76%), but not in the 10 mg BID group (I^2 = 45%).

Headache

As in Fig. S8 (Supporting Information), both subgroups were associated with a significant increase in headache when

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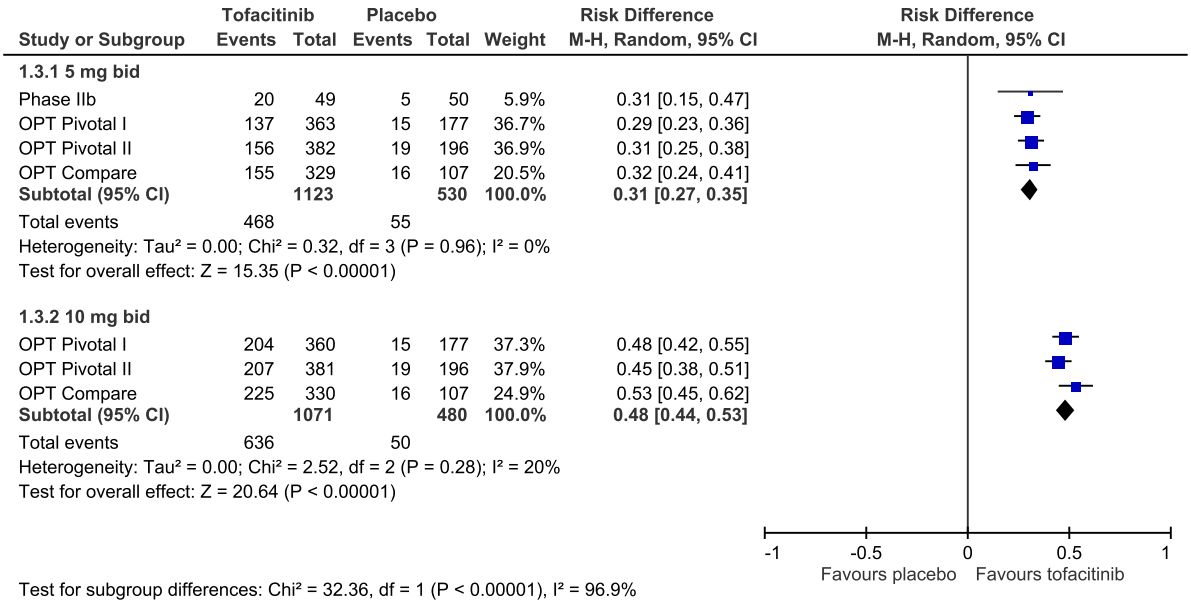

Figure 5 Efficacy of tofacitinib in achieving Physician's Global Assessment of 'clear' or 'almost clear' (0/1).

compared to placebo [5 mg BID: RD 0.03 (95% CI 0.00–0.05), NNH 33 (95% CI 25 to ∞); 10 mg BID: RD 0.03 (95% CI 0.00–0.05), NNH 33 (95% CI 25 to ∞)]. No significant statistical heterogeneity was observed in both comparisons (5 mg BID: I^2 = 14%; 10 mg BID: I^2 = 0%).

Long-term AEs and SAEs Only the pooled data of two RCTs were available.[33] As it is unethical to continue only placebo after the primary endpoint, there were only data on the 5 mg BID and 10 mg BID groups. The two groups did not differ in long-term AEs up to week 52 (567/886 vs. 595/884, P = 0.14). Also, the 5 mg BID and 10 mg BID groups did not differ in long-term SAEs (46/886 vs. 48/884, P = 0.82).

Discussion

This study shows both tofacitinib 5 mg BID and 10 mg BID regimens are effective in treating psoriasis, with the latter more effective than the former. Around one-third of the 5 mg BID group and half of the 10 mg BID group achieved PASI 75 and PGA 0/1 at week 12–16. Nearly one-fifth of the 5 mg BID group and over one-third of the 10 mg BID group achieved PASI 90 at week 12–16. About one-fourth of the 5 mg BID group and one-third of the 10 mg BID group achieved DLQI 0/1, representing no or minimal impairment of life quality. In addition to the improvement of plaque lesions identified in the present study, another post hoc analysis of two included RCTs found tofacitinib also effective in treating nail psoriasis.[34] Furthermore, tofacitinib has been found effective in treating psoriatic arthritis as well.[35]

Our study found tofacitinib well tolerated in treating psoriasis. Only the 10 mg BID group was associated with 9% increase in short-term AEs. Neither the 5 mg BID nor 10 BID groups were related to short-term SAEs. Specifically, both the tofacitinib 5 mg BID and 10 BID groups were associated with a 2% increase in minor adverse events (upper respiratory tract infection and hypercholesterolaemia). A recent study revealed that despite the association with a very small increase in the cholesterol levels, the effects of tofacitinib on lipid-related indicators and circulating cardiovascular risk biomarkers do not suggest an increased cardiovascular risk.[36] We also found no dose–response relation between tofacitinib and long-term AEs or SAEs.

A limitation of this study was the limited number of Asian participants in the included RCTs. Asians occupied from 5% to 13% of the participants.[29–31] However, a Japanese RCT confirmed the efficacy and safety of tofacitinib in treating moderate-to-severe plaque psoriasis.[35] The efficacy of tofacitinib in this Japanese RCT was better than the included RCTs, with 62.8% and 72.7% of participants on 5 and 10 mg BID achieving PASI 75, 37.2% and 54.5% achieving PASI 90 and 67.4% and 68.2% achieving PGA 0/1 at week 16, respectively.[35] The higher efficacy in Japanese patients might be attributed their lower mean bodyweight (67 kg) than those of included RCTs (83–93 kg).[29–31,35] An analysis has confirmed bodyweight affects the efficacy of tofacitinib in treating psoriasis.[37]

Except for Russia, tofacitinib is not approved for use in treating psoriasis elsewhere.[16] However, the evidence up to date supports the efficacy and safety of tofacitinib in treating moderate-to-severe plaque psoriasis. Thus, tofacitinib may be a treatment

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option for moderate-to-severe plaque psoriasis that is unresponsive to other therapies and patients who are intolerable to other therapies or prefer oral medications.

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Supporting information

Additional Supporting Information may be found in the online version of this article:

Table S1. Search strategy
Fig. S1. Efficacy of tofacitinib in achieving a Dermatology Life Quality Index score of 0 or 1.
Fig. S2. Short-term adverse events of tofacitinib.
Fig. S3. Short-term serious adverse events of tofacitinib.
Fig. S4. Safety of tofacitinib as to nasopharyngitis.
Fig. S5. Safety of tofacitinib as to upper respiratory tract infection.
Fig. S6. Safety of tofacitinib as to hypercholesterolemia.
Fig. S7. Safety of tofacitinib as to creatinine phosphokinase elevation.
Fig. S8. Safety of tofacitinib as to headache.

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